

Estimating Infectious Disease Importation Risk during the 2026 FIFA World Cup in the United States, June–July 2026

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The 2026 FIFA World Cup

Key facts

- **48 nations** — largest World Cup in history
- **3 co-host countries:** USA, Mexico, Canada
- **16 venues** across 11 US cities, 2 Canadian, 3 Mexican
- Matches from **June 11 – July 19, 2026**
- Expected **~85 million** international arrivals to the US in June 2026 alone (NTTO forecast)
- Tickets sold to fans from all 48 qualified nations

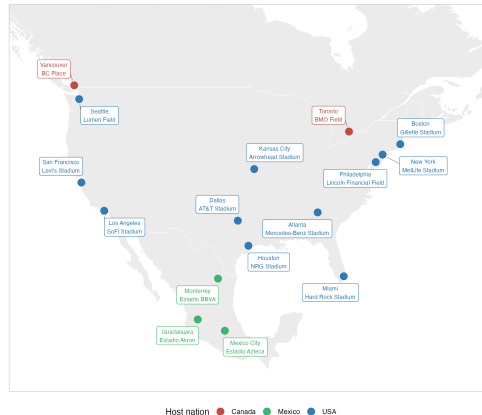
Why does this matter?

Mass gathering events accelerate the **importation** of infectious diseases into host cities. Early quantification helps public health authorities prioritise **surveillance** and **preparedness** by venue.

Venue cities — 16 stadiums across North America

FIFA World Cup 2026 — Venue Stadiums

16 venues across USA (11), Canada (2), and Mexico (3)



Blue: 11 US venues **Red:** 2 Canadian venues **Green:** 3 Mexican venues. Suburb names mapped to metro areas (e.g. East Rutherford → New York).

Scientific question

*What is the probability that at least one case of dengue, influenza, malaria, measles, or pertussis is **imported** into each WC host city during June 2026 — and which source countries drive that risk?*

Five diseases studied

- **Dengue** — endemic in Latin America, SE Asia, Africa; strongly seasonal
- **Influenza** — Southern Hemisphere *winter peak* coincides with June tournament window
- **Malaria** — sub-Saharan Africa, SE Asia, Middle East
- **Measles** — resurgent globally; under-vaccinated pockets

Three nested models

- ① **Baseline** — COR June 2024 $\times$ T-100 routing, no WC adjustment
- ② **WC-adjusted** — 2026 country-level travel surge
- ③ **Schedule-driven** — fan routing by match schedule

The Poisson importation framework

We model disease importation as a **Poisson process**.

If $\Lambda_{v,d}$ is the *expected* number of imported cases of disease d arriving in city v , the probability of at least one importation is:

$$P(X_{v,d} \geq 1) = 1 - e^{-\Lambda_{v,d}} \quad (\text{Poisson})$$

where $\Lambda_{v,d} = \sum_c \lambda_{c,v,d}$ sums over all source countries c .

Each country-level contribution has the same structure across all three models:

$$\lambda_{c,v,d} = \underbrace{N_{c,v}}_{\text{arrivals}} \times \underbrace{I_{c,d}}_{\text{per-traveller infection prob.}} \times \underbrace{p_d}_{\text{travel while infectious}}$$

Disease parameters & uncertainty propagation

Both ρ_d and p_d span wide literature ranges — we sample both simultaneously.

$$I_{c,d} = \rho_d \cdot \frac{C_{c,d}}{P_c}$$

- ρ_d : under-reporting factor
- p_d : probability of travelling while infectious

Disease	ρ_d range	p_d range
Dengue	0.06–0.26	0.30–0.70
Influenza	0.033–0.20	0.30–0.70
Malaria	0.10–0.35	0.10–0.50
Measles	0.40–0.80	0.02–0.10
Pertussis	0.01–0.10	0.50–0.90

Monte Carlo propagation

5,000 independent draws:

$$\rho_d \sim \text{Uniform}(\rho_{\min}, \rho_{\max})$$

$$p_d \sim \text{Uniform}(p_{\min}, p_{\max})$$

Since $\Lambda \propto \rho_d \cdot p_d$, each draw is a scalar rescaling of Λ — *no recomputation needed*.

Output: **MC median + 95% uncertainty interval**

- Dengue & influenza: CI skews **upward** — risk may be understated
- Pertussis: CI skews **downward** — central value is a conservative upper bound

ρ_d sources: Bhatt et al. 2013 (dengue); Iuliano et al. 2018 (influenza); WHO WMR 2024 (malaria); Simons et al. 2012 (measles); McLaughlin et al. 2016

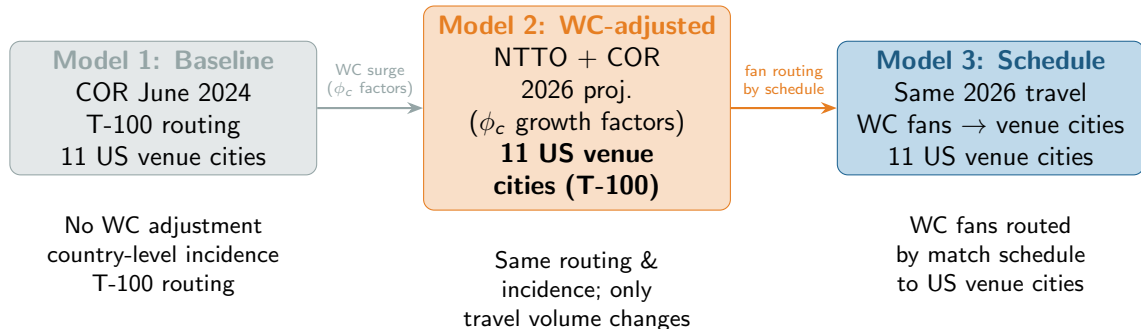
Travel data

- **BTS T-100 International Segment:** nonstop flight segments at individual US airports (June 2023–2025) — country-level routing fractions for all **11 US venue cities**; used in all 3 models
- **COR I-94 data** (CBP): monthly arrivals by country of residence — June 2024 used as baseline (no WC growth for Model 1)
- **NTTO 2026 Forecast** (trade.gov): official 2026 projections for 12 top source markets

Disease data

- **Dengue:** WHO Global Dengue Surveillance (monthly, accessed Dec 2025)
- **Influenza:** WHO FluNet / GISRS global sentinel surveillance (ISO weeks 22–26, 2023–2025)
- **Malaria:** WHO Global Malaria Programme National Unit Data (2024 annual)
- **Measles:** WHO Immunization Data portal (annual, accessed Sep 2025)
- **Pertussis:** WHO Immunization Data portal (annual, accessed Dec 2025)

Three-model hierarchy



M1 → M2: isolates the WC travel surge (ϕ_c growth factors applied to COR 2024 base volumes).

M2 → M3: isolates schedule-based fan routing to US venue cities.

Model 1 — Baseline: equations

Why T-100 routing for the baseline? T-100 covers all **11 US venue cities** at country level, making it the only source that provides a consistent spatial grid across all three models.

Baseline arrivals (COR June 2024, no WC growth):

$$N_{c,v}^{\text{baseline}} = N_{c,\text{Jun 2024}}^{\text{COR}} \cdot f_{c,v}^{\text{T100}} \quad (\text{baseline arrivals})$$

$f_{c,v}^{\text{T100}}$: T-100 routing fraction (mean June 2023–2025), same in all three models, covering **all 11 US venue cities**.

Country-level incidence (shared across all 3 models):

$$I_{c,d} = \rho_d \cdot \frac{C_{c,d}}{P_c} \quad (\text{incidence})$$

Importation intensity:

$$\lambda_{c,v,d} = N_{c,v}^{\text{baseline}} \cdot I_{c,d} \cdot p_d \quad \Lambda_{v,d} = \sum_c \lambda_{c,v,d} \quad (\text{baseline } \lambda)$$

Model 2 — WC-adjusted: travel volume construction

Tiered approach to 2026 travel volume:

Tier 1 — NTTO top-12 markets (Australia, Brazil, Canada, China, France, Germany, India, Italy, Japan, Mexico, South Korea, UK)

Growth factor $\phi_c = V_{c,2026}/V_{c,2024}$ from official NTTO Forecast Report. Already includes WC effect — *do not* add WC visitors on top.

Remaining countries (COR I-94, June 2024)

- WC-qualified: $\phi_{WC} = 85,017/72,390 = \mathbf{1.174}$ (aggregate NTTO WC uplift)
- Non-qualified: $\phi_{base} = \mathbf{1.077}$ — data-driven from COR + T-100 June 2023–2025

$$N_{c, \text{Jun } 2026} = N_{c, \text{Jun } 2024}^{\text{COR}} \cdot \phi_c \quad N_{c,v}^{2026} = N_{c, \text{Jun } 2026} \cdot f_{c,v}^{\text{T100}}$$

Model 2 — WC-adjusted: incidence and λ

Country-level incidence (more precise than regional):

$$I_{c,d} = \rho_d \cdot \frac{C_{c,d}}{P_c} \quad (\text{country incidence})$$

Importation intensity:

$$\lambda_{c,v,d} = N_{c,v}^{2026} \cdot I_{c,d} \cdot p_d \quad \Lambda_{v,d} = \sum_c \lambda_{c,v,d} \quad (\text{WC-adj } \lambda)$$

What changes from Model 1:

- Travel volume reflects the WC surge (ϕ_c growth factors applied to COR June 2024 base volumes)
- Routing fractions, city coverage, disease incidence, ρ_d and p_d are **unchanged** from Model 1

⇒ M1 → M2 isolates the WC travel surge alone

Model 3 — Schedule-driven: the core idea

Problem with Models 1 & 2

T-100 routing fractions reflect habitual tourist flows; WC fans travel to the specific cities where their team plays.

Reality: fans from Brazil go to *Houston* (where Brazil plays), not to Boston or Philadelphia.

Solution: separate the WC-specific travel increment from background tourism, and route fans using the match schedule.

Travel decomposition

$$N_c^{WC} = N_{c,24}^{COR} \cdot \max(0, \phi_c - \phi_{base})$$

$$N_c^{bg} = N_{c,Jun\ 2026} - N_c^{WC}$$

For WC-qualified countries:
WC increment $\approx 8.3\%$ of projected arrivals

$$\frac{1.174 - 1.077}{1.174} \approx 0.083$$

Model 3 — Schedule-driven: routing equations

Schedule-based WC fan routing fraction:

$$\omega_{c,v} = \frac{g_{c,v}}{G_c} \quad (\text{schedule routing})$$

$g_{c,v}$ = games played by country c 's team at venue city v ; G_c = total known group-stage games.

Arrival matrices:

$$N_{c,v}^{\text{WC}} = N_c^{\text{WC}} \cdot \omega_{c,v} \quad (\text{WC fans} \rightarrow \text{venue cities})$$

$$N_{c,v}^{\text{bg}} = N_c^{\text{bg}} \cdot f_{c,v}^{\text{T100}} \quad (\text{background} \rightarrow \text{all 11 US cities via T-100})$$

Combined importation intensity for venue city v :

$$\Lambda_{v,d} = \underbrace{\sum_{c \in Q} N_{c,v}^{\text{WC}} \cdot I_{c,d} \cdot p_d}_{\text{WC fans}} + \underbrace{\sum_c N_{c,v}^{\text{bg}} \cdot I_{c,d} \cdot p_d}_{\text{background}} \quad (\text{schedule } \lambda)$$

Model 3 — Venue coverage

City	State
New York	New York
Dallas	Texas
Houston	Texas
Philadelphia	Pennsylvania
Boston	Massachusetts
Los Angeles	California
Atlanta	Georgia
Kansas City	Missouri*
Miami	Florida
San Francisco	California
Seattle	Washington

mainly WC-fan stream

All 11 cities covered by T-100 routing fractions

* Near-zero T-100 fractions;

Key insight

All 11 US venue cities have both WC-fan and background travel streams — T-100 provides routing fractions for all.

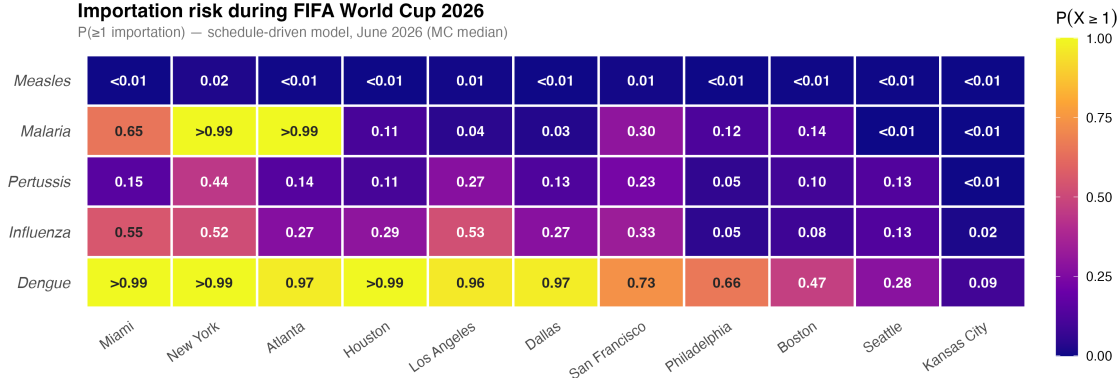
England + Scotland issue:

Both map to “United Kingdom” in COR data. Routing fractions sum to 1 — no double-counting.

Importation risk overview — all five diseases

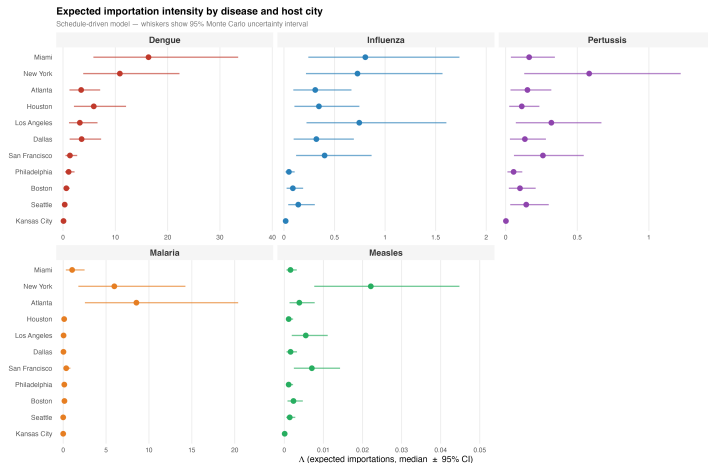
Importation risk during FIFA World Cup 2026

$P(\geq 1 \text{ importation})$ — schedule-driven model, June 2026 (MC median)



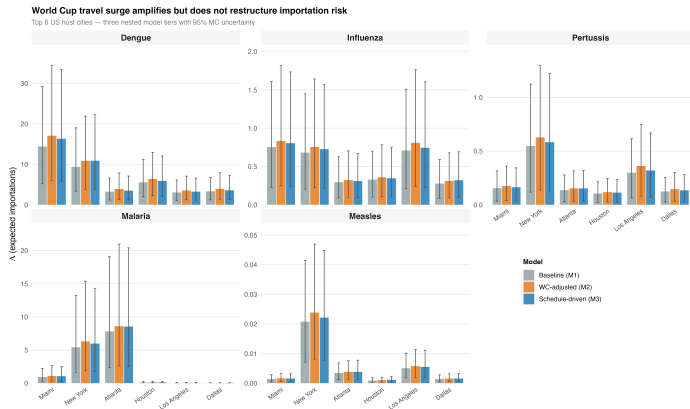
$P(\geq 1)$ under the schedule-driven model (MC median, 5,000 draws). Dengue leads at most venues ($P(\geq 1) > 0.99$ at Miami, New York, and Houston); Atlanta is an exception — malaria exceeds dengue there ($P(\geq 1) > 0.99$ vs. 0.97), driven by direct travel from West and Central African nations. Measles risk is lowest.

Importation intensity with 95% uncertainty intervals



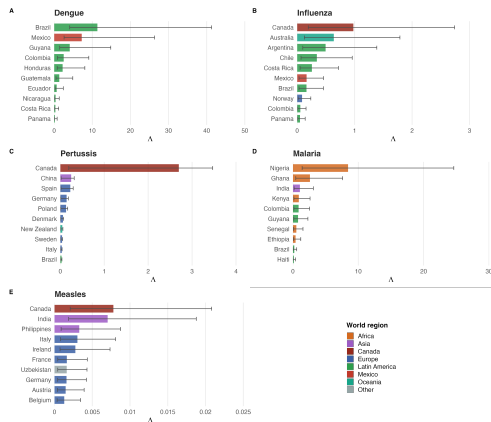
Point = MC median Λ ; whiskers = 95% uncertainty interval (Uniform draws on ρ_d and p_d). Dengue is orders of magnitude above other diseases; pertussis CIs extend downward (central estimate is conservative).

WC travel surge amplifies but does not restructure risk



Grouped bars: Baseline (M1), WC-adjusted (M2), Schedule-driven (M3). Bars = MC median Λ ; whiskers = 95% CI. M1→M2: near-uniform proportional surge across cities. M2→M3: city-specific divergence from schedule-based fan routing.

Top source countries by disease — with uncertainty



Top 10 source countries per disease (schedule-driven, all 11 US cities). Bars = MC median; whiskers = 95% CI. Colours = world region. Latin America: dengue & influenza; Africa: malaria; Europe: pertussis & measles.

Key findings

- 1 **Dengue leads importation risk** at most venue cities ($P(\geq 1) > 0.99$ at Miami, New York, and Houston), robust across the full MC parameter space. **Atlanta is a notable exception**: malaria exceeds dengue there ($P(\geq 1) > 0.99$ vs. 0.97), driven by direct travel from West and Central African nations.
- 2 **Influenza ranks second** at most cities, amplified by coincidence with the Southern Hemisphere winter peak during the June tournament window.
- 3 **The WC travel surge** ($M1 \rightarrow M2$) amplifies risk *proportionally*; city rankings are stable.
- 4 **Schedule-driven routing** ($M2 \rightarrow M3$) concentrates risk at specific US match venues; Kansas City gains meaningful risk almost entirely through the WC-fan stream.
- 5 **Background travel dominates**: the WC-fan increment is $\approx 8.3\%$ of projected arrivals for qualified countries.
- 6 **Country rankings are robust**: the top three contributors per disease are stable across the full MC parameter range.

What this model can (and cannot) offer

What it can offer

- A **transparent, reproducible baseline** for importation risk — built entirely from public data
- A natural **planning timeline**: M1 once cities are known; M2 when projections arrive; M3 once the schedule is confirmed
- Country $\times$ city risk breakdowns that can **inform** which cities and source streams to prioritise for surveillance
- A **quantified uncertainty range** (MC 95% CI) — shows where estimates are robust and where they are not

What it cannot replace

- Local context: vector presence, vaccination coverage, healthcare capacity
- Real-time surveillance — data lags mean this is a *planning* tool, not an operational alert system
- Onward transmission risk after importation
- Event-specific travel data (ticket sales, booking records)

The outputs are one transparent, reproducible input to preparedness planning — not a substitute for local epidemiological judgement.

Limitations

Parameter uncertainty

- **Monte Carlo uncertainty integrated:** 5,000 Uniform draws on (ρ_d, p_d) — 95% intervals reported for all main estimates
- Dengue/influenza CIs skew upward (risk may be understated); pertussis CIs skew downward (conservative upper bound)

Travel data coverage

- T-100 covers all 11 US venue cities across all models
- T-100 covers nonstop segments only; connecting passengers appear at entry city

Model scope

- Importation only — no secondary transmission
- Knockout-round games excluded (TBD opponents)
- Fan routing assumes proportional allocation by games; actual behaviour may differ

Data recency

- Disease data through late 2025; current outbreak situations may differ
- NTTD projections pre-date the event; actual arrival volumes will vary

Data that would most improve the model

Travel volume

- **Flight bookings** (OAG, Amadeus, IATA PaxIS) — direct country-to-city routing for June 2026
- **FIFA ticket sales** by country — most direct measure of WC fan travel; replaces NTTO decomposition
- **Previous WC entry records** (Brazil 2014, Russia 2018, Qatar 2022) — empirical validation of fan routing

Disease burden

- **Real-time surveillance** (ProMED, WHO DON, ECDC CDTR) — catch outbreaks after the 2025 data freeze
- **Age-stratified incidence** — travelers skew working-age; refines $I_{c,d}$
- **Vaccination coverage** (WHO/UNICEF) — adjust effective measles/pertussis incidence for traveler immunity

Local transmission

- **Ae. aegypti surveillance** (CDC ArboNET) — extend to local dengue outbreak probability
- **Hospital capacity** (AHA survey) — contextualise risk for health-system pre-positioning
- **Mobility data** (Meta, Google) — validate schedule-based fan routing empirically

Next steps

- ➊ **Incorporate knockout rounds** — the group stage bracket is now being resolved; update the schedule-driven model with actual match assignments as they are confirmed
- ➋ **Real-time surveillance linkage** — compare model predictions against actual case notifications from CDC and partner health agencies as the tournament progresses
- ➌ **Expand disease scope** — chikungunya (active 2025–2026 outbreak in Brazil & Argentina) and mumps are the highest-priority additions; both have accessible public data
- ➍ **Domestic routing extension** — use DOT enplanement data to capture importation pressure at cities receiving onward domestic connections from T-100 gateway cities
- ➎ **Extended routing data** — booking-level data (OAG, Amadeus) to refine city-level routing fractions beyond T-100 segment data

Thank you

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Code and data:

`https://github.com/jdiestra1977/FIFA\_worldCup\_2026\_risk`

Preprint:

`https://www.medrxiv.org/content/10.64898/2026.06.03.26354828v1`